

Impacts of Groundwater Nitrates on the UC Davis Waterway Algae Blooms



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Background

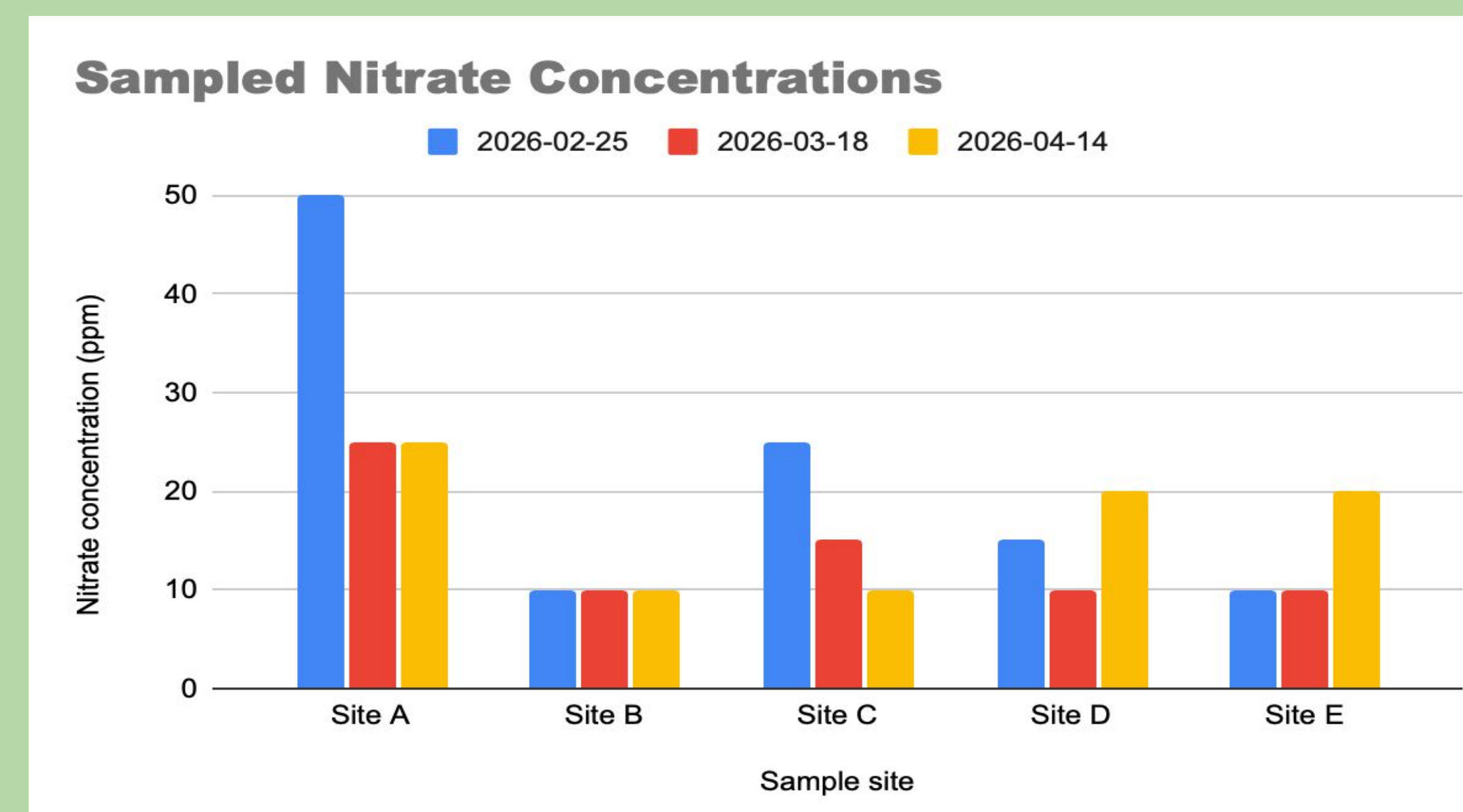
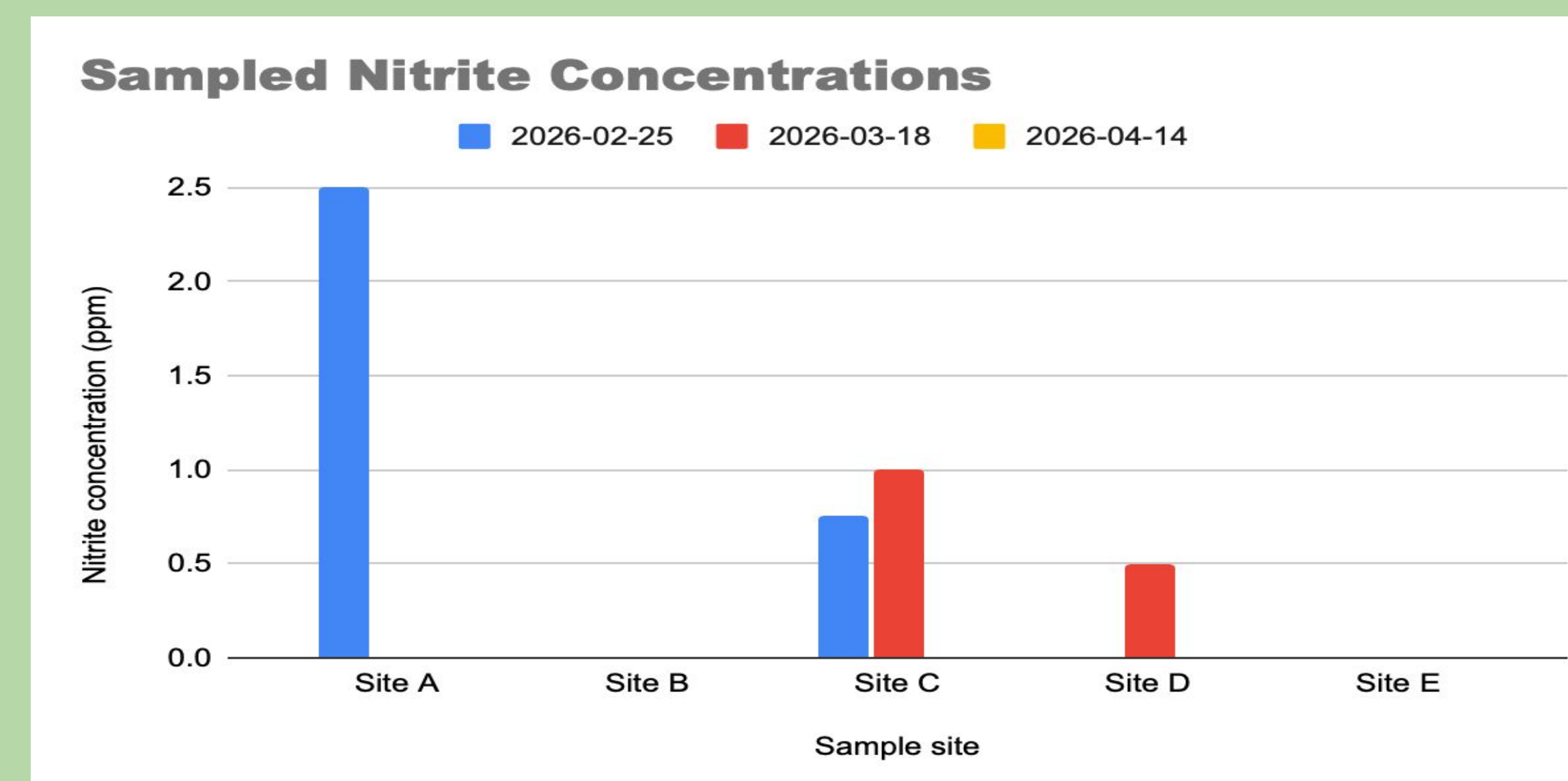
- A Cyanobacteria algae bloom was observed during spring 2025 in Putah Creek on the UC Davis campus [2]
- Cyanobacteria can have harmful effects on both humans and natural ecosystems [2]
- Algae blooms are driven by high concentrations of Nitrogen [2]
- Nitrate and Nitrite are Nitrogen compounds found in groundwater that, alone, have harmful impacts on ecosystems and human health [1]
- We seek to use measured Nitrate and Nitrite concentrations to predict where algae blooms will occur in spring 2026

Hypothesis

We predict that areas with higher Nitrogen (Nitrate/Nitrite) concentrations will be the site of springtime algae blooms.

Results

- No clear trends in Nitrate concentrations between dates
- Site A had the highest concentration of Nitrate during each sample day
- Nitrite concentrations were largely negligible aside from Site A on 2/25
- Only algae bloom that was observed was between Sites A and B on 4/14



Conclusions

Lessons Learned:

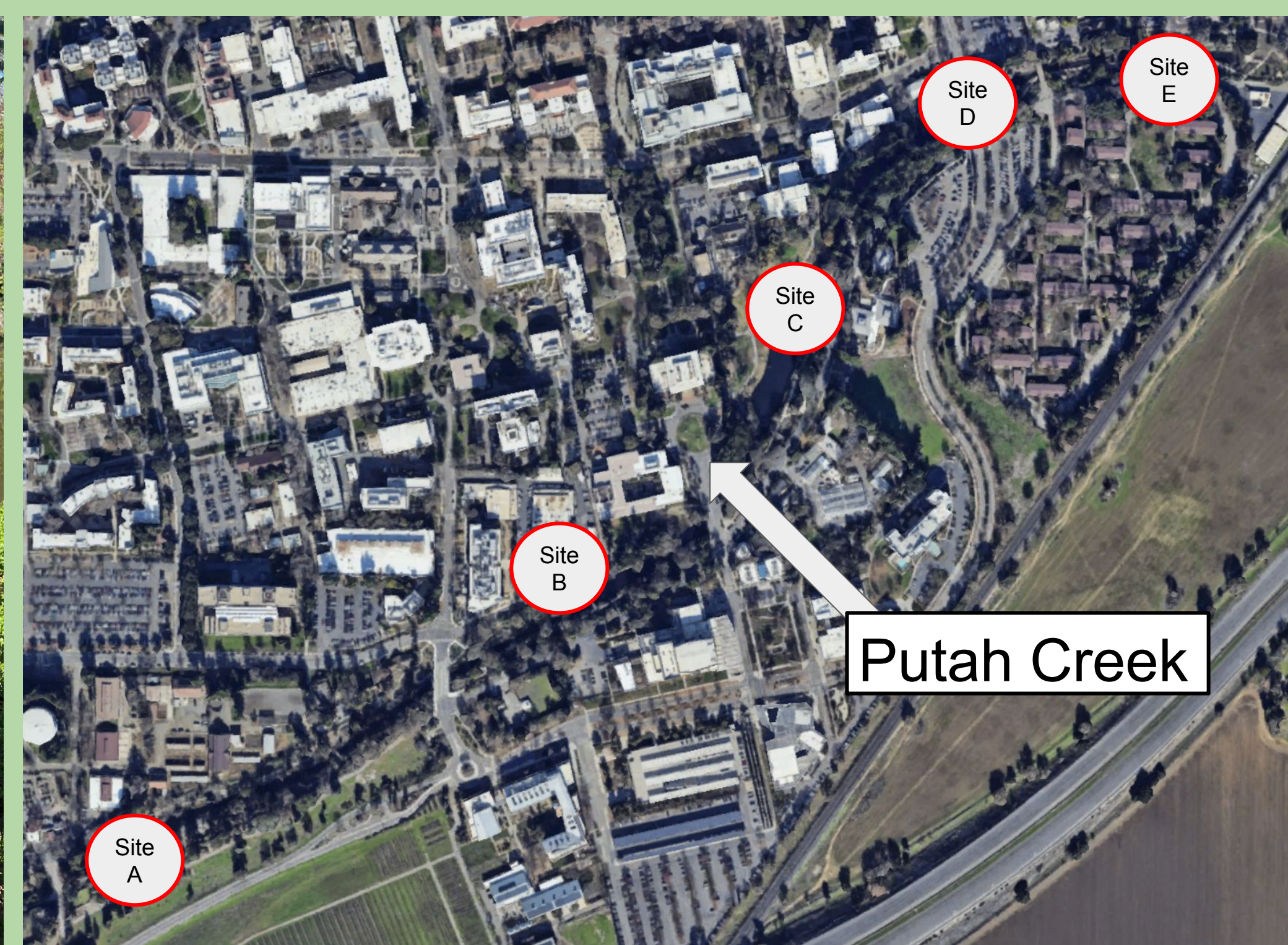
- The observed bloom near Site A, which had the highest Nitrate concentrations, suggesting that our hypothesis may be correct
- The lower concentrations measured at Site B remain to be explained given that the bloom was near this site
- We hypothesize that abnormally low surface Nitrate concentration is the result of subsurface algae that can grow into a surface bloom
- We acknowledge that weather, testing strip accuracy, sampling technique, and human error could all contribute to uncertainty in our data
- Data collection and analysis skills were developed throughout the course of this project

Next Steps:

- Analyze water samples under a microscope to better quantify algae population
- Continue with a higher sampling frequency
- Expanding sample domain to other areas of Putah Creek

Methods

- Nitrogen samples were taken at 5 sites along Putah Creek at the UC Davis Arboretum
- Data was collected using Bartovation Nitrite/Nitrate dip testing sticks
- Samples were collected at three times at approximately monthly intervals
- Algae blooms were qualitatively monitored during our field sample days



References

- [1] California State Water Resources Control Board. *Groundwater Fact Sheet Nitrate*. 2017.
- [2] Mata, Scout. "What's up with the Algae?" *UC Davis Arboretum and Public Garden*, 29 Apr. 2025.